Answers

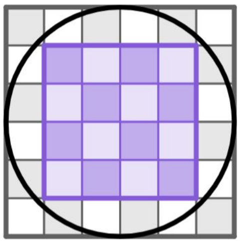
Worksheet

Area of a circle

Task A: Estimating areas of circles

1) For each diagram: i) find the number of complete squares inside the circle;
ii) find the total number of squares;
iii) complete the range for the area of the circle.

a)

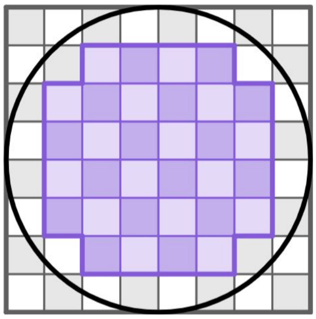


i) number of squares inside = 16

ii) total number of squares = 36

iii) 16 units^2 < area of circle < 36 units^2

b)



i) number of squares inside = 32

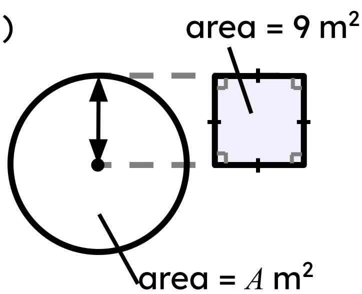
ii) total number of squares = 64

iii) 32 units^2 < area of circle < 64 units^2

2) Use the information provided to fill in the gaps.

The area of each circle is between 2 and 4 times the area of its accompanying square.

a)

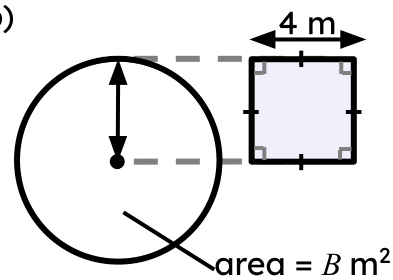


area of 2 squares = 18 m^2

area of 4 squares = 36 m^2

18 < A < 36

b)

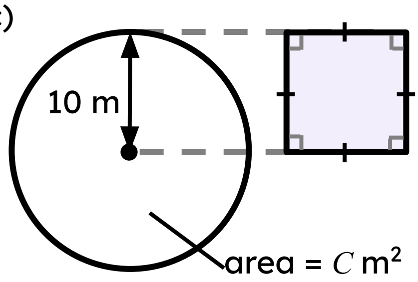


area of 2 squares = 32 m^2

area of 4 squares = 64 m^2

32 < B < 64

c)



area of 2 squares = 200 m^2

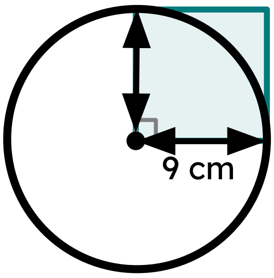
area of 4 squares = 400 m^2

200 < C < 400

Task B: Deriving the formula to calculate areas of circles

1) Each circle overlaps with a square. For each question:
   i) Find the area of the square.
   ii) Find the area of the circle in terms of π.

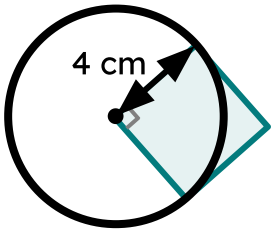
a)



i) area of square = 81 cm^2

ii) area of circle = $81\pi \text{ cm}^2$

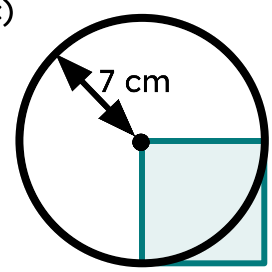
b)



i) area of square = 16 cm^2

ii) area of circle = $16\pi \text{ cm}^2$

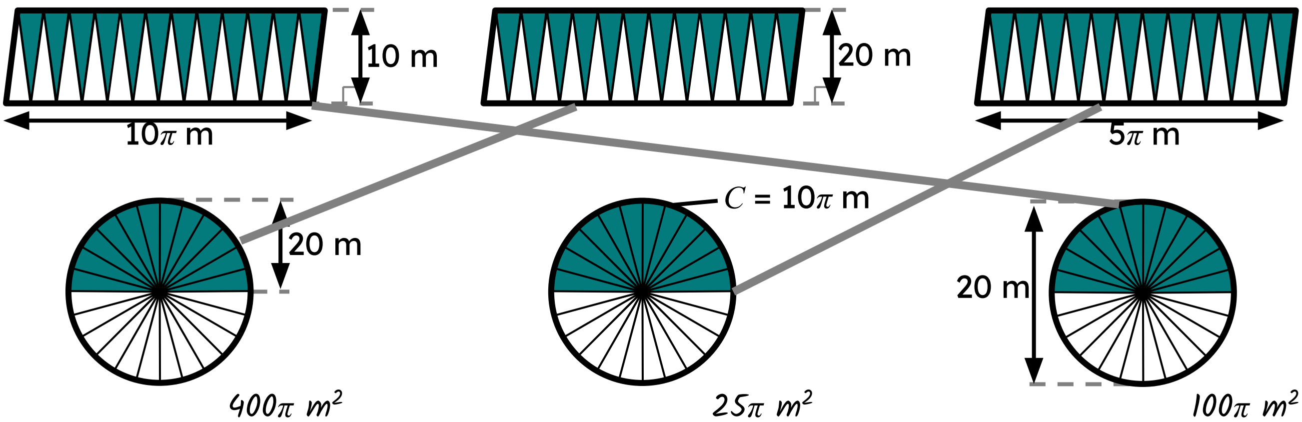
c)



i) area of square = 49 cm^2

ii) area of circle = $49\pi \text{ cm}^2$

2) Each parallelogram is made from the sectors of one of the three circles.
The diagrams are not drawn to scale.



a) Match each parallelogram with the circle of the same approximate area.

b) Find the area of each circle in terms of π.